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Handling you with care

FROM DATA TO INTELLIGENCE

Scaling AIoT Opportunities at Dipped Products PLC (DPL), Kottawa

This report outlines an AIoT roadmap for Dipped Products PLC (DPL), which currently functions as a hybrid operation where significant data is captured but critical decision-making remains largely manual and human-dependent. It recommends a phased transition that prioritizes flexible packing automation and mixing sensors to build a robust IoT foundation before integrating intelligent decision-making layers, with a strong focus on responsible workforce redeployment.

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Executive Summary

This report examines how Dipped Products PLC (DPL), a leading rubber glove manufacturer in the Hayleys Group, can use the Internet of Things (IoT) and Artificial Intelligence (AI) to improve its manufacturing process. It is based on a field visit to the DPL factory in Kottawa, where the production process was observed from raw material handling to final packing.

The visit showed that DPL already collects a significant amount of data through its Enterprise Resource Planning (ERP) system, laboratory records, machine logs, and quality checkpoints. However, much of this data is still used to support human decision-making rather than automated or intelligent responses. Two areas stood out during the visit and are used as case studies throughout this report. The first is compound mixing, where formulation errors may only be identified after the required maturation period and laboratory testing. The second is manual quality control, where worker fatigue can reduce inspection accuracy and increase the risk of defects being missed.

A phased approach is recommended. First, computer vision should be introduced to support manual quality inspection and identify visible defects. This should be followed by automated counting and digital verification in packaging, particularly for high-volume and standardised pack types. The third phase should focus on mixing through inline sensors and automated dosing, while keeping important adjustments under human approval. Once reliable data has been established across these areas, an Artificial Intelligence combined with IoT (AIoT) layer can be introduced to identify patterns, predict potential problems, and provide decision recommendations.

The report also considers the business case, implementation approach, and workforce impact of these changes. The main constraint is not expected to be investment capacity, but the speed of implementation, system integration, and operational risk. Critical safety checks, such as the pinhole and voltage-leak test, should remain deterministic and should not be replaced by probabilistic AI decisions. Automation should also be supported by retraining and role rotation so that affected workers can transition into suitable roles rather than being viewed only as a reduction in headcount. This approach would allow DPL to gradually move from collecting production data to using it for faster, more informed, and predictive decision-making.

1. Introduction

1.1 Purpose and Scope

This report is based on a field visit made by our team to the Dipped Products PLC (DPL) factory in Kottawa. The purpose of the visit was to understand the rubber glove manufacturing process and identify where technologies such as the Internet of Things (IoT) and Artificial Intelligence combined with IoT (AIoT) could be used to improve the process. The report documents the observations made during the visit, highlights stages where the process remains heavily dependent on manual work, and proposes practical improvements based on feasibility, cost, and workforce impact.

1.2 DPL and the Hayleys Group

DPL was established in 1976 and is the oldest and one of the largest plants within its group, which today includes 13 plants producing around 5 million pairs of gloves. DPL is part of the Hayleys Group, one of Sri Lanka's largest and most diversified companies, and is itself listed on the Colombo Stock Exchange. It is recognised as one of the world's largest glove manufacturers, with exports reaching more than 70 countries. The company is supported by its own Research and Development (R&D) team, rubber plantations, and manufacturing plants located in Sri Lanka and Thailand.

1.3 How Rubber Gloves Are Made

The basic glove manufacturing process begins with a hand-shaped mould known as a former, which is dipped into a tank containing liquid rubber, such as natural latex or synthetic materials such as nitrile or neoprene. The former is first heated before being dipped, and the process may be repeated to achieve the required glove thickness. The rubber layer is then dried to form the glove. Once formed, the gloves are subjected to quality checks, testing, and cleaning before being packed and shipped. The raw materials are also tested before being allowed into the factory to ensure that they meet DPL's quality standards.





2. Factory Floor Observations

2.1 Raw Material Sourcing and Incoming Quality Control

Each batch of raw material is tested before use, regardless of supplier data. A colour tag system is used to show the status of each batch. Green indicates passed, blue indicates on hold, and orange indicates not yet checked.

Material that fails specifications may be adjusted and approved through a signed Non-Conformance Report (NCR) or returned to the supplier. For example, a material with a Total Solids Content (TSC) reading of 40.9 was approved following technical review despite being below specification. Imported chemicals and pigments account for approximately 99.9% of usage and are mainly flammable, so they are stored externally. Although largely manual, the process is structured and rule-based, providing a contrast to compound mixing.

2.2 Case Study A: Compound Mixing

Chemicals are combined according to batch card standards recorded in SAP. Quantities can be specified with high precision. For example, a 500 g component may be required within a 6,000 kg batch. Dosing is therefore controlled through SAP and materials are pumped into a pressurised tank.
However, individual chemicals must first be lab-tested, and the mixed compound requires a maturation period of up to 3.5 days for some formulations. Further laboratory testing is then carried out for viscosity, colour, TSC, and pH before production approval.

A formulation difference of around 100 g in a 500 g component can stop production for two to three days. The main limitation is therefore not dosing accuracy but the delay between mixing, maturation, and testing. Errors may only be identified several days after the mixture is prepared.

2.3 Dipping Process and Levels of Automation

Different levels of automation were observed across DPL's 13 plants, established between 1976 and 2026. Older plants rely more heavily on manual monitoring. For example, coagulant immersion may be timed at approximately 18 seconds ±2 seconds, while oven temperatures of 145°C ±2°C may be checked manually up to six times daily. One plant was reported to produce approximately 45,000 gloves per day with a shippable yield of around 98%.



A fully robotic cell was also observed, where formers are heated, dipped, and transferred between robots according to timed cycles. Production can reach approximately 8,500 gloves per hour, although some manual work remains, such as bubble removal.

Human oversight is still required because latex viscosity can vary with weather and seasonal conditions. Defects may also result from raw materials, timing, temperature, contamination, tank vibration, or dipping speed. Process changes therefore continue to require human judgement and technical approval.

2.4 Quality Control Processes

Quality control varies by glove type. Electrical and insulating gloves undergo an automated voltage leak or pinhole test. However, loading, operation, and sorting remain manual.
For most other glove types, quality control is fully manual, with trained staff visually and physically inspecting each glove.

Case Study B: Manual Quality Control and Worker Fatigue

It was reported that inspection accuracy decreases after approximately two hours of continuous work, even among trained workers. This creates a significant quality risk, particularly for medical and other high-risk gloves where critical defects cannot be accepted.

The main gap is the lack of automated inspection for glove types outside the existing electrical testing process. Smart inspection systems could therefore help identify defects that may be missed due to worker fatigue.





2.5 Packing and Limited Automation

Packing was identified by management as an area with significant potential for improvement. Each pack is barcoded to provide approximately two years of traceability, while customer-specific logos and artwork may also be required. A dedicated checker manually verifies each pack against the order sheet.

Automation remains limited mainly because of the high variation in packing requirements. DPL handles approximately 100 pack types, with customers requiring different folds, logo positions, and box formats. A fixed automated system would therefore be difficult to justify unless sufficient flexibility could be achieved.





2.6 Laboratory Operations

Laboratories support most stages of production. Raw materials are tested against DPL standards, while the laboratories also support product development, reverse engineering, and durability testing for properties such as stretch, tear, puncture, and abrasion.

Laboratory results are entered directly into DPL's information systems rather than being printed and transferred manually. This shows that digital data collection is already present in parts of the process and provides a foundation for further IoT and AIoT integration.

3. The Core Gap: Data Exists, but Intelligence Does Not

DPL already collects substantial data through laboratory results, order specifications, machine logs, and quality control records. Much of this is recorded through the Enterprise Resource Planning (ERP) system, although some paper records remain. The main gap is therefore not the availability of data, but how it is used.

When process parameters drift, batches are placed on hold, or defect trends develop, the data is still reviewed by a person before action is taken. Data currently supports human decision-making but does not independently trigger responses.

The recommendations are therefore divided into two stages. First, an Internet of Things (IoT) layer should improve real-time sensing, automated data collection, and data reliability. An Artificial Intelligence combined with IoT (AIoT) layer can then be introduced to analyse this data and support intelligent decisions.

Artificial Intelligence should not be introduced before the underlying data and sensing issues are addressed, as unreliable data could lead to unreliable decisions. A strong IoT foundation is therefore necessary for the AIoT layer.

4. The IoT Layer: Building the Sensing Foundation

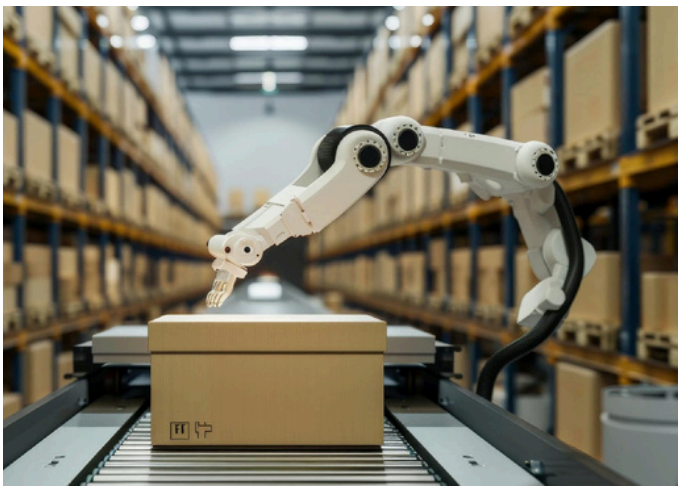
4.1 Revisiting Case Study B: Quality Control and Worker Fatigue

A computer vision system should first be introduced to support manual inspection rather than immediately replace it. Industrial cameras and Artificial Intelligence (AI) models could continuously identify visible defects, reducing the effect of worker fatigue and providing an additional quality check. The CCA-YOLO model, developed for nitrile medical gloves, achieved mean Average Precision (mAP) above 99% for several tear and scratch defects. Commercial systems have also demonstrated high-speed automated glove inspection.



Initially, clear defects could be flagged automatically while uncertain cases remain subject to inspector review. Critical tests such as the pinhole test should continue to use deterministic methods rather than AI-based decisions. This approach would allow the system to be tested using DPL's own production data before wider automation is considered.

4.2 Automated Counting and Packaging Support



The next step should focus on improving the packaging process through automated counting and digital verification. Automated counting systems could be used to ensure that the correct number of gloves is placed in each pack, while barcode scanning could verify the product against the relevant order information.
This would reduce manual counting errors and provide more consistent production records without requiring immediate replacement of the entire packing process.

Such systems are already used in glove manufacturing, where automated counting, cartoning, and barcode or Radio Frequency Identification (RFID) verification are used to improve packing accuracy and traceability.
Given DPL's large number of pack variations, counting and verification should be introduced before full robotic packaging. The system could first be applied to high-volume and standardised products before being expanded to more complex pack types.

4.3 Revisiting Case Study A: Compound Mixing

Once the quality control and packaging improvements are established, the mixing process should be strengthened with inline sensors on maturation and feed tanks. Continuous monitoring of viscosity, pH, and Total Solids Content (TSC) would allow deviations to be detected during the process instead of after several days of maturation and laboratory testing. Industrial refractometers such as the Vaisala Polaris PR53 are already used for continuous concentration monitoring in rubber and latex applications.
For small chemical adjustments, a Human-in-the-Loop (HITL) dosing process is recommended. When an abnormal reading is detected, the system could calculate the required adjustment and send it to a responsible executive for approval. Once approved, automated pumps could deliver the required amount, while the Enterprise Resource Planning (ERP) system records the adjustment and approver.
Similar automated weighing and dosing systems are already used in rubber compounding to improve recipe accuracy and traceability. DPL could therefore build on its existing SAP-based process rather than replacing it.

4.4 Specific Sensor Options for the Mixing Retrofit

Two sensor types are particularly relevant to the mixing process. Optical refractometers such as the Vaisala Polaris can continuously measure concentration through the refractive index of the material, while vibrating-rod viscometers can continuously measure viscosity. This would provide more immediate process information than periodic laboratory testing.
When connected to DPL's existing Programmable Logic Controllers (PLCs) and ERP system, these sensors could identify deviations and recommend adjustments before a batch is fully committed. For example, a drop in coagulant concentration caused by water carryover could trigger an adjustment recommendation, with final approval remaining under human control.
The existing laboratory testing could therefore be retained as a quality safeguard while the IoT system reduces the likelihood of major deviations being discovered only after maturation. The data generated across quality control, packaging, and mixing would then provide the foundation for the predictive AIoT layer proposed in the final phase.

5. The AIoT Layer: Adding Intelligence

5.1 Predictive Decision-Making Across the Production Process

The Artificial Intelligence combined with the Internet of Things (AIoT) layer should be introduced only after reliable data has been established across quality control, packaging, and mixing. At this stage, data from sensors, inspection systems, production records, and the Enterprise Resource Planning (ERP) system could be analysed together to identify patterns and predict potential problems.

For example, historical quality data could be used to identify defect trends across machines and shifts before they become major quality issues. Similarly, mixing data such as viscosity, pH, Total Solids Content (TSC), and temperature could be analysed to identify conditions associated with unstable batches. Machine learning has already been studied for predicting rubber compound properties such as tensile strength, showing potential for predictive applications in this industry.

The same data could also support predictive decisions in packaging and quality control. Instead of responding only after a defect, delay, or process deviation occurs, the system could identify early warning patterns and recommend appropriate action.

5.2 Human-in-the-Loop Decision-Making

The final AIoT layer should not be designed to make every decision independently. A Human-in-the-Loop (HITL) approach should be maintained for decisions that could affect production, quality, or safety.

When an unusual pattern is detected, the system could provide a prediction or recommended action to the responsible employee. The decision could then be reviewed and approved before any significant process adjustment is made. This would allow AI to support technical staff without removing human responsibility from important decisions.

For example, if mixing data indicates that a batch is likely to move outside its required viscosity range, the system could recommend an adjustment based on previous production data. The responsible executive could review the recommendation before automated dosing is allowed.

5.3 A Portfolio of Methods Rather Than One AI System

Different production problems require different technologies. The existing electrical test can identify pinholes affecting electrical insulation, while computer vision can identify visible scratches, tears, or other surface defects. Similarly, mixing sensors provide direct measurements that AI can later use for prediction.

A combination of purpose-built technologies should therefore be used rather than relying on one AI system for the entire production process. AI should be applied where it provides a clear advantage, while physical sensors, deterministic tests, and human inspection should remain where they are more reliable.

5.4 Where Artificial Intelligence Should Not Be Used

AI should not be used as the sole basis for decisions where a guaranteed pass or fail result is required. Machine learning systems operate using probabilities and confidence levels, which are not suitable for critical safety decisions.

The existing pinhole and voltage leak tests should therefore remain deterministic and rule-based, particularly for medical, electrical, and other high-risk gloves. AI can instead be used to support areas such as surface defect detection, process prediction, and early identification of production problems.

Overall, the AIoT layer should act as the final stage of the proposed roadmap. By combining reliable data from quality control, packaging, mixing, and other production stages, DPL could move from simply recording production information to using it for predictive and better-informed decisions.

6. Business Case: Is It Worth the Investment?

6.1 Quality Control

Computer vision should be considered the first major automation investment because it directly addresses the risk of fatigue during manual inspection. Its main benefit would be the addition of a consistent inspection layer rather than the immediate replacement of inspectors.

Industry applications have shown that AI-based vision systems can operate at high production speeds and detect defects that may be difficult to identify consistently through manual inspection. The investment could therefore be justified through improved quality consistency, reduced rework, and lower risk of defective products reaching customers.

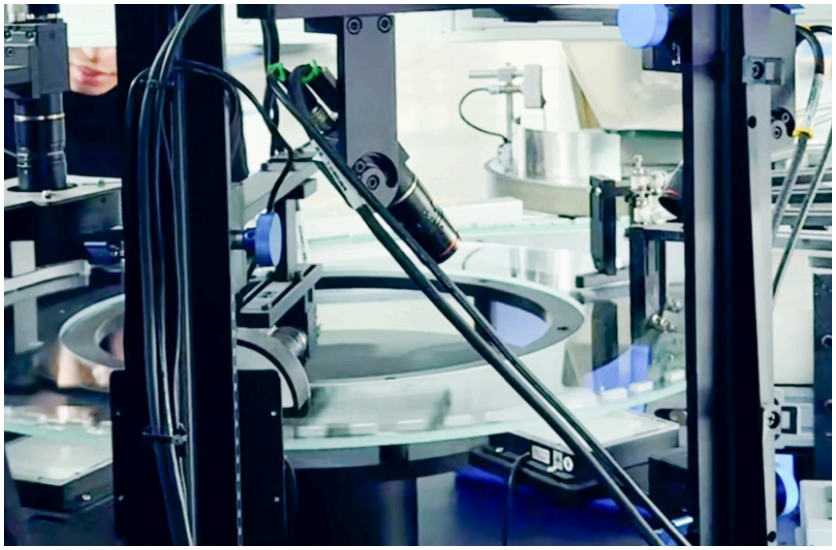
A pilot should first be introduced on a selected production line so that its accuracy, operational impact, and financial benefits can be assessed before wider deployment.

6.2 Packaging and Automated Counting

The next investment should focus on automated counting and digital verification rather than full robotic packaging. DPL handles more than 100 pack types with different folding methods, logos, artwork, and box formats, making complete packaging automation difficult to justify immediately.

Automated counting systems can improve quantity accuracy, while barcode verification can ensure that the correct product is matched with the relevant order. Industry solutions have already combined automated counting, cartoning, and barcode verification in glove manufacturing.

For DPL, counting and digital verification should first be introduced for high-volume and standardised pack types. This would provide measurable improvements without requiring the entire packaging process to be automated at once. Full robotic packaging could be considered later where sufficient standardisation exists.



6.3 Compound Mixing

Investment in mixing instrumentation should follow the packaging improvements because the cost of a failed batch can be significantly higher than the cost of monitoring equipment. DPL's existing laboratory testing prevents unsuitable compounds from reaching production, but problems may only be identified after the maturation period.

Inline sensing would provide earlier warnings and reduce material waste and production delays. Automated dosing could then be introduced to make small adjustments based on sensor readings, with Human-in-the-Loop (HITL) approval retained for important changes.

Industry examples also show that automated dosing and weighing systems can reduce formulation errors while improving traceability. Mixing instrumentation is therefore a suitable investment once the earlier data collection systems have been established.

6.4 Predictive AIoT

The final investment stage should be the development of the Artificial Intelligence combined with the Internet of Things (AIoT) layer. By this stage, reliable data would be available from quality control, packaging, mixing, and other production activities.

This data could be used to identify trends, predict potential failures, and recommend actions before problems affect production. The financial value would therefore come not only from reducing individual errors, but from improving overall process stability and reducing unexpected downtime.

The AIoT system should remain Human-in-the-Loop, particularly where decisions could affect product quality or production. Critical safety checks should continue to use deterministic methods rather than AI-based probability.

6.5 Overall Investment Consideration

A phased approach would reduce the financial and operational risk of introducing automation across the factory. Each stage would generate practical results and better-quality data that can support the following stage.

The main constraint is therefore not expected to be capital availability, but implementation speed, system integration, and workforce transition. Automation should be supported by retraining and role transition so that affected workers can move into areas such as equipment monitoring, quality verification, and maintenance support.

7. Implementation Layer

DPL already has an in-house product development team, and funding was not identified as the main barrier to implementation. The greater priority should therefore be placed on execution speed, system integration, and avoiding disruption to daily production. Implementation should follow the same phased approach recommended in this report, beginning with computer vision, followed by packaging counting, mixing instrumentation, and finally predictive Artificial Intelligence combined with the Internet of Things (AIoT).

A hybrid implementation approach is recommended. DPL's internal team should lead areas requiring detailed knowledge of the factory and its existing systems, while external specialists can be engaged for technical areas such as computer vision, sensor integration, and machine learning. This would allow specialist knowledge to be introduced while operational knowledge remains within DPL.

A pilot-based approach should be followed rather than implementing large-scale systems at once. The computer vision system should first be tested on a selected production line, followed by automated counting for suitable packaging types. Mixing sensors and dosing systems can then be introduced and connected to the existing systems. Once reliable data has been established across these areas, predictive AIoT applications can be developed.

This gradual approach is particularly important because previous automation efforts, including the introduction of robotic arms, were affected by configuration issues that took more than a month to resolve. Each stage should therefore be tested under actual production conditions before wider deployment. External partners should also train DPL's internal team so that the technologies can eventually be managed and improved internally without long-term dependence on vendors.

8. Ethical Considerations and Workforce Transition

The introduction of automation would affect workers currently involved in quality inspection and packaging, particularly contract workers. However, workforce reduction should not be treated as the primary objective of automation. A transition plan should instead be included from the beginning of the implementation process.

Computer vision should initially be used to support quality inspectors rather than replace them. Similarly, packaging automation should be introduced gradually, allowing time for affected workers to be reassigned or retrained. As identified by management, opportunities exist within other areas of the company where contract workers could be rotated as automation is introduced.

Training could be provided for roles involving machine monitoring, quality verification, equipment support, and other activities that continue to require human involvement. Human oversight should also be retained for important AIoT decisions, particularly where product quality or safety could be affected.

This approach would allow DPL to gain the benefits of automation while retaining the practical knowledge and experience of its workforce. Automation should therefore be treated as a transition in roles and responsibilities rather than simply a reduction in jobs.

9. Conclusion and Phased Recommendations

DPL has already demonstrated its ability to adopt advanced automation, particularly through its robotic dipping operations. However, mixing, quality control, and packing still depend heavily on manual work. The main opportunity is therefore not simply to introduce more technology, but to connect the data already being generated with systems that can support faster and more consistent decisions.

A four-phase roadmap is recommended:

Phase 1: Computer Vision for Quality Control. A computer vision system should first be introduced to support manual inspection by detecting surface-level defects. Human inspectors should remain involved in reviewing uncertain cases, while critical tests such as the pinhole test should remain fully deterministic.

Phase 2: Automated Counting and Packaging Support. Automated counting and digital verification should then be introduced to reduce manual counting errors and improve packing accuracy. This can initially be applied to the most standardised and high-volume pack types before wider packaging automation is considered.

Phase 3: Automated Mixing and Dosing. Inline sensors should be introduced to monitor mixing conditions, with automated dosing systems used to prepare small chemical adjustments. The existing SAP system can be connected to these processes so that recipe accuracy and traceability are maintained, while final adjustments remain subject to human approval.

Phase 4: Predictive AIoT Decision-Making. Once reliable data is being collected across quality control, packing, mixing, and other production stages, Artificial Intelligence combined with IoT (AIoT) can be introduced to identify patterns and predict potential problems. Human-in-the-Loop (HITL) controls should remain in place so that important decisions are reviewed and authorised by responsible personnel.

The overall objective is to **move DPL from a factory that mainly collects data to one that uses its data to act earlier and more intelligently.** At every stage, safety and workforce protection should remain priorities. Critical safety decisions should not be based solely on AI probabilities, and automation should be accompanied by retraining and role transition plans for affected workers.

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